Week 3: Signal Conditioning & GPIO Protection - Theory Notes

Signal Conditioning and GPIO Protection

These notes provide detailed calculations, component selection guidance, and practical design considerations for interfacing real-world signals with your MCU.

Voltage Divider Design

Basic Formula

A voltage divider scales an input voltage by a fixed ratio:

$$V_{out} = V_{in} \times \frac{R2}{R1 + R2}$$

The divider ratio (attenuation factor):

$$k = \frac{V_{out}}{V_{in}} = \frac{R2}{R1 + R2}$$

Design Procedure

Step 1: Determine the required ratio

$$k = \frac{V_{out,max}}{V_{in,max}}$$

For 0-10V input to 0-3.3V output: k = 0.33

Step 2: Choose a practical R2 value (typically 1kΩ to 10kΩ)

Step 3: Calculate R1

$$R1 = R2 \times \left(\frac{1}{k} - 1\right)$$

Example: k = 0.33, R2 = 10kΩ

$$R1 = 10k \times \left(\frac{1}{0.33} - 1\right) = 10k \times 2.03 = 20.3k\Omega$$

Use 20kΩ standard value.

Loading Effects

When a load (like an ADC) is connected to the divider output, it appears in parallel with R2:

$$R2_{eff} = R2 \parallel R_{load} = \frac{R2 \times R_{load}}{R2 + R_{load}}$$

This changes the divider ratio:

$$V_{out,loaded} = V_{in} \times \frac{R2_{eff}}{R1 + R2_{eff}}$$

STM32 ADC Input Impedance:

The ADC input impedance depends on the sampling time setting:

Sampling Time	Approximate Input Impedance
3 cycles	~6 kΩ
15 cycles	~17 kΩ

Sampling Time	Approximate Input Impedance
84 cycles	~50 kΩ
480 cycles	~200 kΩ

Recommendation: Use longer sampling times (84+ cycles) when using voltage dividers, or use lower divider resistance.

Source Impedance Requirement

The STM32 datasheet specifies maximum recommended source impedance for the ADC. The voltage divider's Thevenin equivalent resistance is:

$$R_{source} = R1 \parallel R2 = \frac{R1 \times R2}{R1 + R2}$$

Example: R1 = 20kΩ, R2 = 10kΩ

$$R_{source} = \frac{20k \times 10k}{20k + 10k} = 6.67k\Omega$$

For accurate readings, ensure Rsource < recommended maximum for your sampling time.

When Does Settling Error Actually Appear?

The settling error depends on how much C_{ADC} must charge between consecutive samples — not just on R_{AIN} and sampling time.

Single channel, DC (repeated reads): C_{ADC} retains its charge between conversions. After the first sample, subsequent reads start with the capacitor already at approximately the correct voltage. The incremental charging needed is negligible, so even 3 sampling cycles produce accurate results regardless of R_{AIN}. This is why a simple bench test with a DC voltage divider may show no error at all — even with 200kΩ/100kΩ resistors and 3-cycle sampling.

After a channel switch (multi-channel scanning): This is the primary real-world trigger. When the ADC switches from one channel to another, C_{ADC} is still charged to the *previous* channel's voltage. The new channel must charge C_{ADC} across the full voltage difference through R_{AIN} in a single sampling window. With high source impedance and short sampling time, C_{ADC} cannot settle.

Fast-changing signals (square wave): A square wave on a single channel can also trigger the error, but only on the few samples that land immediately after an edge. Since C_{ADC} tracks the signal between edges, most samples are already settled. The effect is visible as occasional spikes rather than a consistent offset.

Scenario	3 cycles + 66.7kΩ	480 cycles + 66.7kΩ
Single channel, DC	Accurate (C_{ADC} already settled)	Accurate
After channel switch (0V → 3.3V)	Reads significantly low	Settles correctly
Square wave (single channel)	Occasional spikes near edges	Settles correctly

To observe the effect experimentally: Read a grounded channel (PA1 tied to GND) immediately before reading the divider channel (PA0). This forces C_{ADC} to swing from 0V to the divider voltage on every sample. Compare 3-cycle vs 480-cycle readings — the 3-cycle values will be consistently and visibly lower than the 480-cycle values with the high-impedance divider.

Practical Design Table

Input Range	R1	R2	Ratio	Rsource	Accuracy with 50kΩ load
0-5V	1kΩ	2kΩ	0.67	667Ω	$<$ 1% error
0-10V	2kΩ	1kΩ	0.33	667Ω	$<$ 2% error
0-12V	2.7kΩ	1kΩ	0.27	730Ω	$<$ 2% error
0-24V	6.8kΩ	1kΩ	0.128	872Ω	$<$ 2% error

RC Low-Pass Filter Design

Transfer Function

For a first-order RC low-pass filter:

$$H(f) = \frac{1}{1 + j\frac{f}{f_c}}$$

Magnitude response:

$$|H(f)| = \frac{1}{\sqrt{1 + \left(\frac{f}{f_c}\right)^2}}$$

Cutoff frequency:

$$f_c = \frac{1}{2\pi RC}$$

Design Procedure

Step 1: Determine cutoff frequency

- Choose fc above the highest signal frequency you want to keep
- Choose fc below the noise frequencies you want to remove
- Common choice: fc = 10× signal bandwidth

Step 2: Choose capacitor value (common values: 100nF, 1µF)

Step 3: Calculate resistor value

$$R = \frac{1}{2\pi f_c C}$$

Filter Response at Key Frequencies

Frequency	Attenuation
0.1 × fc	-0.04 dB (99.5%)
0.5 × fc	-1.0 dB (89%)
fc	-3.0 dB (71%)
2 × fc	-7.0 dB (45%)
10 × fc	-20 dB (10%)

Combined Divider and Filter

When combining a voltage divider with an RC filter, the divider's lower resistor can serve as the filter resistor:

```
        R1         R2
Vin   [ ]    [ ]     Vout
```

```
              GND          C

                      GND
```

In this configuration:

- Voltage divider: R1 and R2
- RC filter: R2 and C

$$f_c = \frac{1}{2\pi R2 \times C}$$

Design Example:

Goal: Scale 0-12V to 0-3.3V with 100 Hz cutoff

1. Divider ratio: 3.3/12 = 0.275
2. Choose R2 = 10kΩ, calculate R1 = 26.4kΩ (use 27kΩ)
3. For fc = 100 Hz: C = 1/(2 × 10k × 100) = 159nF (use 150nF)

Filter Settling Time

After a step change in input, the filter output takes time to settle:

$$\tau = RC$$

Time to reach percentage of final value:

Time	Output Level
1	63%
2	86%
3	95%
5	99%

Example: R = 10kΩ, C = 100nF

$$\tau = 10k \times 100n = 1ms$$

Output reaches 99% of final value in 5ms.

GPIO Protection Fundamentals

STM32 Absolute Maximum Ratings

From the STM32F401 datasheet:

Parameter	Value
VDD	-0.3V to 4.0V
GPIO voltage (standard pins)	-0.3V to VDD + 0.3V
GPIO voltage (5V tolerant pins)	-0.3V to 5.5V
GPIO current (sink/source)	±25 mA max
Total GPIO current	120 mA max

5V Tolerant Pins: Check datasheet - marked as "FT" in pinout table. These pins have modified input structure but still limited to ~5.5V.

Internal ESD Protection

STM32 GPIOs have internal protection diodes:

- Clamp to VDD: ~0.6V forward drop
- Clamp to GND: ~0.6V forward drop
- Current handling: Typically < 5mA sustained

The internal diodes are for ESD protection during handling, not for sustained overvoltage!

Series Resistor Protection

A series resistor limits current through the internal protection diodes:

$$I_{diode} = \frac{V_{signal} - V_{clamp}}{R_s}$$

Where Vclamp   VDD + 0.6V for overvoltage, or -0.6V for undervoltage.

Design Rule: Choose Rs so Idiode < 5mA for worst-case overvoltage.

Example: Protect against 12V signal

$$R_s > \frac{12V - 3.9V}{5mA} = 1.62k\Omega$$

Use 2.2kΩ for margin.

Effect of Series Resistor on Digital Signals

Series resistance creates an RC time constant with input capacitance:

$$\tau = R_s \times C_{in}$$

STM32 GPIO input capacitance: ~5pF typical

Example: Rs = 10kΩ

$$\tau = 10k \times 5pF = 50ns$$

Rise time 2.2 = 110ns

This limits maximum frequency for digital signals. For 1 MHz signals, keep Rs < 1kΩ.

External Clamping Circuits

Schottky Diode Clamp

```
            VDD

               D1 (BAT54)


Signal  [Rs]    GPIO

               D2 (BAT54)


            GND
```

Operation:

- D1 clamps positive overvoltage to VDD + Vf
- D2 clamps negative voltage to -Vf

Schottky vs Silicon Diodes:

Parameter	Schottky (BAT54)	Silicon (1N4148)
Forward voltage	0.2-0.4V	0.6-0.8V
Reverse recovery	< 5ns	~4ns
Clamp level (VDD=3.3V)	3.5-3.7V	3.9-4.1V

Recommendation: Use Schottky for 3.3V systems.

Zener Diode Clamp

```
Signal   [Rs]    GPIO

                   Zener (3.3V)


                 GND
```

Advantages:

- Clamps to fixed voltage regardless of VDD
- Works even when MCU power is off
- Simple, single component

Disadvantages:

- Zener voltage tolerance (±5% typical)
- Higher capacitance than Schottky
- Some leakage below breakdown

Part Selection: Use 3.3V Zener (e.g., BZX84C3V3) for 3.3V systems.

TVS Diode Protection

TVS (Transient Voltage Suppressor) diodes are designed for ESD and transient protection:

Key Parameters:

Parameter	Description
Standoff voltage (Vrwm)	Normal operating voltage
Breakdown voltage (Vbr)	Voltage where clamping begins
Clamping voltage (Vc)	Voltage during surge
Peak pulse current (Ipp)	Maximum surge current

Example Part: PESD3V3S1BL

- Vrwm = 3.3V
- Vbr = 6.4V (typ)
- Vc = 15V @ 1A
- Capacitance = 50pF

Use Case: Place at connector for external signals to protect against ESD from user contact.

Level Shifting

5V to 3.3V (Input)

Option 1: Voltage Divider

Simple, works for DC and low-frequency signals.

$$R1 = 1k\Omega, R2 = 2k\Omega \rightarrow 5V \times \frac{2k}{3k} = 3.33V$$

Option 2: Series Resistor Only (5V tolerant pins)

For 5V tolerant GPIO pins, just use series resistor for protection.

Option 3: Dedicated Level Shifter

For high-speed signals, use a dedicated IC like 74LVC1T45.

3.3V to 5V (Output)

Check Compatibility First:

Many 5V CMOS/TTL inputs recognize 3.3V as logic HIGH:

Logic Family	VIH (min)	3.3V Compatible?
5V CMOS (CD4000)	3.5V	No
5V HC/HCT	2.0V	Yes
5V TTL	2.0V	Yes
5V LVTTL	2.0V	Yes

If Not Compatible:

Option 1: Open-drain with pull-up

```
3.3V GPIO      5V Device Input
(open-drain)
           [10k] Pull-up to 5V

           5V
```

Configure GPIO as open-drain output. Pull-up to 5V provides high level.

Option 2: BSS138 MOSFET Level Shifter

```
3.3V                    5V

  [10k]   [10k]

   LV       HV

          S
            BSS138
          G

         GND
```

This is bidirectional - works in both directions automatically.

Option 3: Level Shifter IC

For multiple signals or high speed, use dedicated ICs:

- 74LVC245: 8-bit, unidirectional
- TXB0108: 8-bit, auto-direction
- TXS0102: 2-bit, auto-direction

Practical Design Examples

Example 1: 0-10V Sensor to ADC

Requirements:

- Input: 0-10V analog sensor
- Output: 0-3.3V to STM32 ADC
- Bandwidth: DC to 10 Hz
- Protection: Handle up to 15V fault condition

Design:

```
Sensor      Rs       R1         R2       ADC
output    [2.2k] [4.7k]    [2.2k]   input


                              1µF


                           GND
```

Analysis:

1. Divider ratio: 2.2k / (4.7k + 2.2k) = 0.32 $\rightarrow$ 10V $\times$ 0.32 = 3.2V
2. Filter cutoff: 1 / (2 $\times$ 2.2k $\times$ 1µF) = 72 Hz
3. Protection: 2.2kΩ series limits current at 15V to (15-3.9)/2.2k = 5mA

Example 2: 5V Digital Input

Requirements:

- Input: 5V logic signal from Arduino
- Output: 3.3V compatible for STM32 GPIO
- Speed: Up to 100 kHz

Design:

```
5V signal  [1k]   [2k]  GND

                          GPIO
```

Analysis:

1. Output voltage: 5V $\times$ 2k/(1k+2k) = 3.33V
2. Time constant: (1k 2k) $\times$ 5pF = 3.3ns $\rightarrow$ supports > 10 MHz

Example 3: Protected Digital Input with ESD

Requirements:

- Input: External connector (user accessible)
- Signal: 0-5V digital
- Protection: ESD + overvoltage

Design:

```
Connector     Rs      TVS               GPIO
       [1k]

                 GND      GND
```

Use PESD5V0S1BL (5V TVS) for ESD protection.

Common Mistakes and Solutions

Mistake	Consequence	Solution
Divider resistors too high	ADC loading error, noise pickup	Keep total $< 10\text{k}\Omega$
No series resistor	Internal diode damage	Always add Rs 1kΩ
Wrong Zener voltage	Clamps too high or too low	Match to VDD
TVS at wrong location	Doesn't protect connector	Place at connector, not at MCU
Ignoring capacitance	Slow signal edges	Use smaller Rs for fast signals
No protection when MCU off	Current flows through clamp to VDD	Use Zener to GND

Quick Reference

Voltage Divider Formula

$$V_{out} = V_{in} \times \frac{R2}{R1 + R2}$$

RC Filter Cutoff

$$f_c = \frac{1}{2\pi RC}$$

Protection Current

$$I_{max} = \frac{V_{signal} - V_{clamp}}{R_s}$$

Standard Resistor Values (E24 series)

1.0, 1.1, 1.2, 1.3, 1.5, 1.6, 1.8, 2.0, 2.2, 2.4, 2.7, 3.0, 3.3, 3.6, 3.9, 4.3, 4.7, 5.1, 5.6, 6.2, 6.8, 7.5, 8.2, 9.1 × 10^n

Common Capacitor Values

10pF, 22pF, 47pF, 100pF, 220pF, 470pF, 1nF, 2.2nF, 4.7nF, 10nF, 22nF, 47nF, 100nF, 220nF, 470nF, 1µF, 2.2µF, 4.7µF, 10µF